



DAFFODIL INTERNATIONAL UNIVERSITY

Department of Computing and Information System (CIS)

Web Development Essentials

Semester Assignment

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Assignment Tasks

TASK-1

a. Explain the CSS Box Model and describe how it can be used to create well-structured sections such as the Alumni Directory and Success Stories on the University Alumni Portal.

Answer: The CSS Box Model is an important concept in web design. It describes how every HTML element is treated as a rectangular box. The box consists of four main parts:

- **Content:** The actual text, image, or other information inside an element.
- **Padding:** The space between the content and the border.
- **Border:** The line surrounding the element.
- **Margin:** The outside space between one element and another.

For example, in my Alumni Portal, each alumni profile is displayed inside a card. Padding gives space between the profile information and the card border, while margin creates space between different alumni cards. Border and border-radius also help make the cards visually attractive.

The Box Model helps maintain proper spacing and alignment. Therefore, it is useful for creating organized sections such as the **Alumni Directory** and **Success Stories**.

b. Explain the purpose and importance of a University Alumni Portal. Describe the roles of HTML, CSS, and Bootstrap in developing the University Alumni Portal.

Answer: An Alumni Portal is important because it can:

1. Connect alumni with each other.
2. Provide career and job opportunities.
3. Share university news and announcements.
4. Promote alumni success stories.
5. Provide information about upcoming events.
6. Allow alumni to register and communicate with the association.

Role of HTML:

HTML is used to create the basic structure of the website. In my project, HTML is used to create headings, paragraphs, images, buttons, forms, cards, navigation sections, and other webpage elements.

Role of CSS:

CSS is used to design and style the HTML elements. It controls colors, fonts, spacing, borders, backgrounds, layouts, hover effects, and responsiveness.

Role of Bootstrap:

Bootstrap is a CSS framework that helps create responsive and professional websites quickly. In my Alumni Portal, Bootstrap is used for responsive grids, cards, buttons, forms, containers, and the image carousel.

Therefore, HTML creates the structure, CSS provides the design, and Bootstrap helps with responsive and reusable components.

TASK-2

a. Construct the HTML structure for the Home Page of the University Alumni Portal including the university logo, welcome message, banner image, and a "Join Alumni Network" button.

Answer: The Home Page of my Alumni Portal contains the university logo, navigation menu, a banner section, welcome message, and a call-to-action button.

A simplified structure is:

```
<div id="topheader">
  <div id="email">
    <div class="icon"><i class="ri-mail-fill"></i></div>
    <div class="texticon">Email : 253-16-019@diu.edu.bd</div>
  </div>
  <div id="phone">
    <div class="icon"><i class="ri-phone-fill"></i></div>
    <div class="texticon">Hotline : +8801758046685</div>
  </div>
  <div id="login">
    <div id="firstcolor" class="log">Login</div>
    <div id="secondcolor" class="log"><a href="#rashed" style="text-decoration: none;">Join Alumni
      Network</a></div>
  </div>
</div>

<div id="header">
  <div class="logo"></div>
  <div id="optiontext">
    <h2>Home</h2>
    <h2>Events</h2>
    <h2>Blogs</h2>
    <h2>Notice</h2>
    <h2>Community</h2>
  </div>
  <div class="contacttext">
```

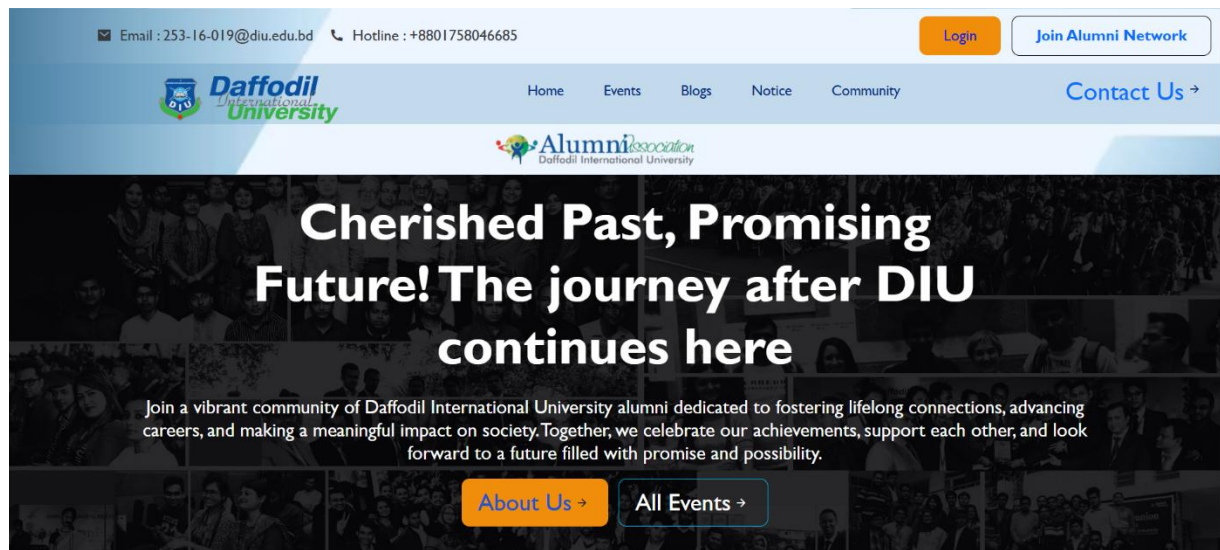
```

    <h2><a href="#rashed" style="text-decoration: none;">Contact Us</a></h2>
    <div class="icon"><i class="ri-arrow-right-line"></i></div>
  </div>
</div>

<div id="page1">
  <div id="box">
    <h1>
      Cherished Past, Promising
      Future! The journey after DIU
      continues here
    </h1>
    <h4>
      Join a vibrant community of Daffodil International University alumni dedicated to fostering
lifelong
      connections, advancing careers, and making a meaningful impact on society. Together, we
celebrate
      our achievements, support each other, and look forward to a future filled with promise and
      possibility.
    </h4>
    <div id="btns">
      <div id="btn1">
        <h2><a href="#bigbox">About Us</a></h2>
        <div class="icon"><i class="ri-arrow-right-line"></i></div>
      </div>
      <div id="btn2">
        <h2>All Events</h2>
        <div class="icon"><i class="ri-arrow-right-line"></i></div>
      </div>
    </div>
  </div>
</div>
  </div>

```

Output :



b. Design the About Alumni Association section using CSS to improve typography, spacing, colors, and visual appearance.

Answer: The About Alumni Association section explains the purpose and activities of the Alumni Association. I used CSS to make this section visually attractive and easy to read.

First, The **HTML** can be written as:

```
<div id="bigbox">
  <div id="topic1" class="topic">
    <h2>About Alumni Association</h2>
  </div>
  <div id="topic2" class="topic">
    <h2>Why you should join us</h2>
  </div>
  <div id="topic3" class="topic">
    <div class="littlebox">
      
      <h2>Lifelong Bond and Networking for Success</h2>
      <p>Joining the Alumni Association opens doors to a vast network of accomplished
individuals,
      providing unparalleled opportunities for personal and professional growth.</p>
    </div>
    <div class="littlebox">
      
      <h2>Thriving as One and Nurturing Growth Together</h2>
      <p>A dedicated platform for networking, career support, job opportunities, mentorship
      programs, and industry insights to propel your professional journey forward.</p>
    </div>
    <div class="littlebox">
      
      <h2>Create Impact and Give Back to the Community</h2>
      <p>Contribute to the alma mater's legacy by participating in philanthropic initiatives,
      giving back to the community, and creating a impact for future generations.</p>
    </div>
  </div>
</div>
```

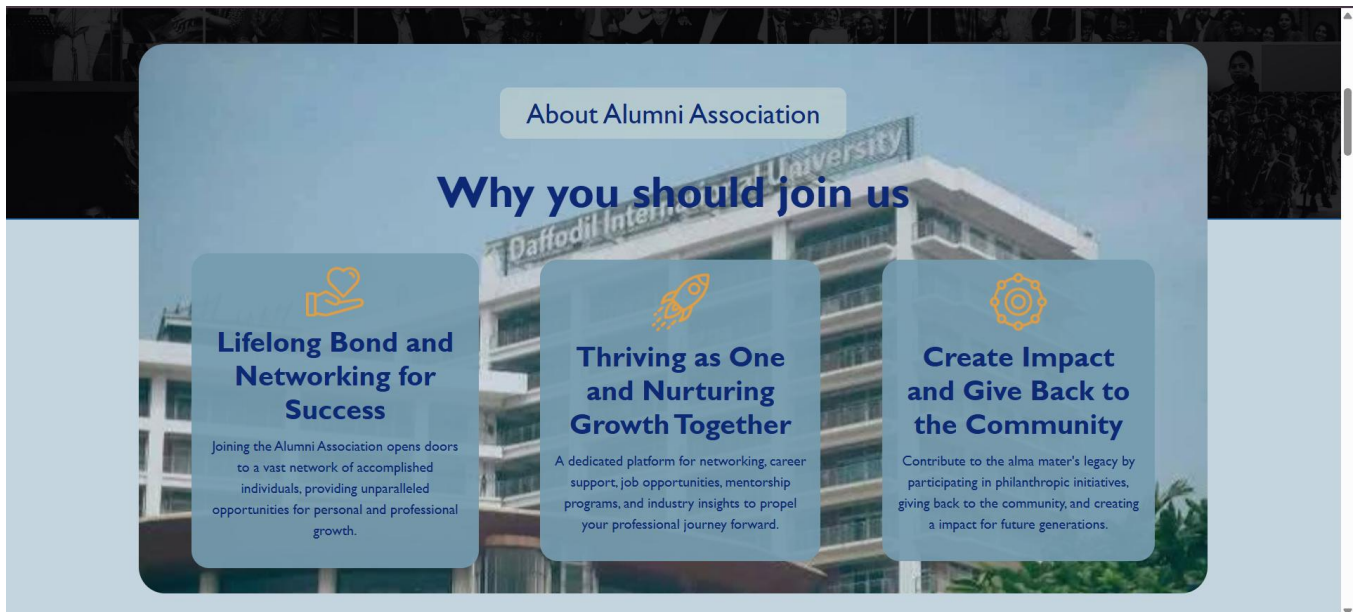
Now,

The **CSS** can be written as:

```
#bigbox{
    height: 90%;
    width: 80%;
    background-image:
url(534550410_122162106728461695_5277020272853931808_n.jpg);
    background-size: cover;
    background-repeat: no-repeat;
    position: relative;
    top:8%;
    left: 10%;
    text-align: center;
    padding: 30px;
    border-radius: 30px;
}
.topic{
    margin: auto;
    display: flex;
    align-items: center;
    justify-content: center;
}
#topic1 h2{
    width: auto;
    padding: 10px 30px;
    border-radius: 10px;
    background-color: rgba(245, 245, 220, 0.326);
    margin: 20px;
    color: rgb(14, 41, 118);
    font-weight: lighter;
}
#topic2 h2{
    color: rgb(14, 41, 118);
    margin: 10px;
    font-size: 50px;
    font-weight: 700;
    font-weight: bolder;
}
#topic3{
    display: flex;
    gap: 10px;
}
.littlebox{
    color: rgb(14, 41, 118);
}
```

```
margin: 30px;
width: auto;
background-color: rgba(117, 156, 175, 0.93);
height: auto;
padding: 15px;
border-radius: 20px;
cursor: pointer;
transition: transform 0.3s ease;
}
.littlebox h2{
margin-top: 10px;
font-weight: bolder;
}
.littlebox p{
margin-top: 10px;
}
.littlebox:hover{
transform: scale(1.1);
}
```

Output:



c. Use CSS Flexbox or CSS Grid to organize the Alumni Directory or Success Stories section in a structured and visually balanced layout.

Answer: For the Alumni Directory, CSS Grid is a good choice because the alumni profiles can be arranged into rows and columns.

The **CSS** can be written as:

```
.dictionary {
  width: 30%;
  margin: 0 auto 20px auto;
  text-align: center;
}

.dictionary h3 {
  padding: 12px 20px;
  margin: 0;
  background-color: rgba(66, 151, 249, 0.7);
  border-radius: 12px;
  font-size: 32px;
  font-weight: 500;
  color: #111;
}

.alumni-container {
  width: 90%;
  margin: 0 auto;
  display: grid;
  grid-template-columns: repeat(3, 1fr);
  gap: 25px;
}

.alumni-card {
  background-color: #eff0f2;
  border-radius: 6px;
  overflow: hidden;
  border: 1px solid #d8dce0;

  display: flex;
  flex-direction: column;

  transition: 0.3s ease;
}

.alumni-card:hover {
  transform: translateY(-7px);
  box-shadow: 0 8px 20px rgba(0, 0, 0, 0.15);
}
```



```
}  
.alumni-image {  
  width: 100%;  
  height: 350px;  
  background-color: white;  
  padding: 5px;  
}  
.alumni-image img {  
  width: 100%;  
  height: 100%;  
  object-fit: cover;  
  display: block;  
}  
.alumni-content {  
  padding: 15px;  
  flex: 1;  
}  
.alumni-content h2 {  
  margin-bottom: 15px;  
  color: #b47601;  
  font-size: 20px;  
  font-weight: 600;  
}  
.alumni-content p {  
  margin: 0;  
  color: #111;  
  font-size: 15px;  
  line-height: 1.5;  
}  
.alumni-footer {  
  margin: 0 20px;  
  padding: 12px 0;  
  border-top: 1px solid #c9c9c9;  
  color: #777;  
  font-size: 15px;  
}  
.see-all {  
  width: 100%;  
  display: flex;  
  justify-content: flex-end;  
  margin-top: -40px;  
}  
.see-all a {  
  color: #0066cc;  
  font-size: 18px;
```

```
text-decoration: underline;
}
.see-all a:hover {
    color: #003d7a;
}
```

Output:



d. Implement responsive design techniques using CSS Media Queries so that the University Alumni Portal adapts properly to mobile, tablet, and desktop screens.

Answer:

Responsive design means making a website suitable for different screen sizes. I used CSS Media Queries to change the layout according to the device width.

```
@media (max-width:1024px) {
    #topheader {
        margin-left: 20px;
        font-size: large;
    }
    #login {
        margin-left: auto;
    }
    #secondcolor {
        font-size: medium;
    }
    #optiontext {
        margin-left: auto;
    }
    #optiontext h2 {
        margin-left: 20px;
    }
}
```

```
}
.contacttext {
    margin-left: auto;
}
#box h1 {
    font-size: 50px;
    width: auto;
    padding: 10px 200px 5px 200px;
}
#box h4 {
    width: auto;
    padding: 2px 100px 5px 100px;
    font-size: medium;
}
#btns{
    gap: 8px;
}
#bigbox{
    width: 88%;
    left: 6%;
}
#topic1 h2{
    margin: 25px;
}
#topic2 h2{
    margin: 5px;
    font-size: 40px;
}
#topic3{
    gap: 10px;
}
.littlebox{
    margin: auto;
    padding: 10px;
    width: auto;
    height: auto;
}
#mediumbox{
    top: 40%;
}
.log{
    padding: 8px 20px;
}
#carouselExampleIndicators{
    padding: 15px;
}
.card-title{
    font-size: 20px;
}
```

```
#page4 {
    padding: 60px 4%;
}

.dictionary {
    width: 45%;
    margin-bottom: 80px;
}

.dictionary h3 {
    font-size: 28px;
}

.alumni-container {
    gap: 18px;
}

.alumni-image {
    height: 380px;
}

.alumni-content h2 {
    font-size: 21px;
}

.alumni-content p {
    font-size: 15px;
}
#page8{
    border-radius: 60px 60px 0 0;
}

}
@media (max-width:768px) {
    #main{
        background-image: none;
    }
    #topheader{

        align-items: center;
        justify-content: center;
        height: auto;
        margin: 10px;
        font-size: 15px;
    }
    #email,#phone,#login{
        margin: 5px 0;
    }
    #header{
```

```
        height: auto;
        padding: 15px;
    }
    .logo{
        margin: auto;
    }
    .logo img{
        width: 160px;
    }
    #optiontext{
        flex-wrap: wrap;
        justify-content: center;
        margin: 15px 0;
    }
    #optiontext h2{
        margin: 10px;
        font-size: 16px;
    }
    .contacttext{
        margin: auto;
        padding: 4px;
    }
    #box h1{
        font-size: 38px;
        padding: 30px 50px 5px 50px;
    }
    #box h4{
        font-size: 16px;
        padding: 10px;
    }
    #btns{
        flex-direction: column;
        gap: 15px;
    }

    #btn1,#btn2{
        width: 250px;
    }
    #btn1 h2,#btn2 h2{
        font-size: 24px;
    }
    #bigbox{
        width: 95%;
        left: 2.5%;
        padding: 20px;
    }
    #topic1 h2{
        font-size: 24px;
        margin: 10px;
```

```
}
#topic2 h2{
    font-size: 32px;
}
.littlebox{
    width: 100%;
    margin: 10px 0;
}
.littlebox h2{
    font-size: 20px;
}
#mediumbox{
    width: 95%;
    left: 2.5%;
    top: 20%;
    align-items: end;
}
#vcImg{
    width: 90%;
    justify-content: center;
}
#vcDetails{
    width: 100%;
    margin-top: 20px;
}
#wlcm{
    font-size: 24px;
    text-align: center;
}
#wlcmDetails{
    font-size: 14px;
    text-align: left;
}
.dictionary{
    width: 80%;
    left: 10%;
}
.card-group{
    padding: 20px;
}
.carousel-inner img{
    height: 350px;
}
.gallery-img{
    height: 180px;
}
.log{
    margin: 5px;
    padding: 8px 20px;
}
```

```
        font-size: 14px;
    }
    #firstcolor,#secondcolor{
        padding: 8px 20px;
    }
    #vcImg img{
        width: 100%;
    }
    #carouselExampleIndicators{
        padding: 10px;
    }
    .card-title{
        font-size: 18px;
    }
    .event-box{
        padding: 15px;
    }
    .social-btn{
        padding: 6px 12px;
        font-size: 14px;
    }
    #page4 {
        padding: 50px 5%;
    }
    .dictionary {
        width: 80%;
        margin-bottom: 50px;
    }
    .dictionary h3 {
        font-size: 25px;
    }
    .alumni-container {
        gap: 10px;
    }
    .alumni-image {
        height: auto;
    }
    .alumni-image img {
        height: auto;
        max-height: 500px;
        object-fit: contain;
    }
    .alumni-content h2 {
        font-size: 18px;
    }
    .alumni-content p {
        font-size: 10px;
    }
    .see-all {
```

```
        margin-top: 20px;
    }
    #page8{
        border-radius: 40px 40px 0 0;
    }
}
@media (max-width:400px) {
    #topheader{
        font-size: 13px;
        text-align: center;
    }
    #email,#phone{
        display: none;
    }
    #login{
        gap: 20px;
    }
    #header{
        flex-direction: column;
        padding: 10px;
    }
    #logo2{
        width: 25%;
        object-fit: contain;
    }
    .logo img{
        width: 140px;
    }
    #optiontext{
        align-items: center;
    }
    #optiontext h2{
        margin: 6px;
        font-size: 16px;
    }
    .contacttext h2{
        font-size: 15px;
    }
    #box h1{
        font-size: 28px;
        padding: 15px;
    }
    #box h4{
        font-size: 14px;
        padding: 10px;
    }
    #btn1,#btn2{
        width: 90%;
        height: 50px;
    }
}
```

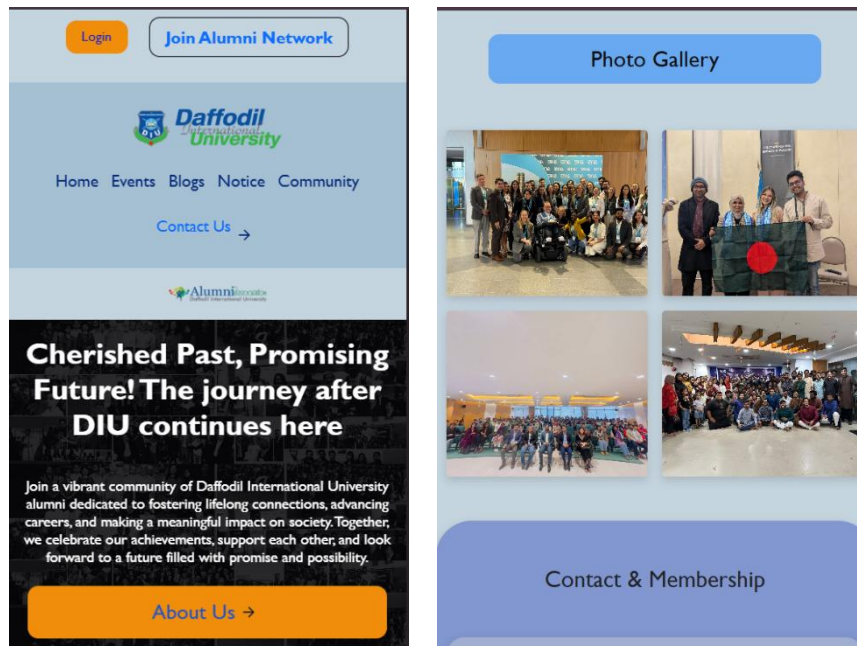


```
}  
#btn1 h2,#btn2 h2{  
    font-size: 20px;  
}  
#bigbox{  
    width: 98%;  
    left: 1%;  
    padding: 15px;  
}  
#topic1 h2{  
    font-size: 18px;  
}  
#topic2 h2{  
    font-size: 24px;  
}  
.littlebox{  
    padding: 15px;  
}  
.littlebox h2{  
    font-size: 15px;  
}  
.littlebox p{  
    font-size: 10px;  
}  
#mediumbox{  
    width: 98%;  
    left: 1%;  
}  
#vcImg{  
    width: 50%;  
    object-fit: contain;  
}  
#wlcm{  
    font-size: 20px;  
}  
#wlcmDetails{  
    font-size: 12px;  
}  
.card-group{  
    padding: 5px;  
}  
.carousel-inner img{  
    height: 220px;  
}  
.gallery-img{  
    height: 150px;  
}  
.contact-box,.form-box{  
    padding: 5px;
```

```
}
.contact-box h4, .form-box h4{
    font-size: 10px;
}
.log{
    margin: 4px auto;
    padding: 6px 15px;
    font-size: 13px;
}
#firstcolor, #secondcolor{
    padding: 6px 15px;
}
#carouselExampleIndicators{
    padding: 5px;
}
.card-title{
    font-size: 16px;
}
.event-box{
    padding: 12px;
}
.event-box h5{
    font-size: 16px;
}
.social-btn{
    padding: 5px 10px;
    font-size: 12px;
    margin: 3px 3px 0 0;
}
.form-control{
    font-size: 14px;
}
#page4 {
    padding: 40px 15px;
}
.dictionary {
    width: 95%;
    margin-bottom: 40px;
}
.dictionary h3 {
    font-size: 21px;
    padding: 10px;
}
.alumni-content {
    padding: 5px;
}
.alumni-content h2 {
    font-size: 10px;
}
```

```
.alumni-content p {
    font-size: 9px;
    line-height: 1.6;
}
.alumni-footer {
    margin: 0 15px;
    font-size: 13px;
}
#page8{
    border-radius: 25px 25px 0 0;
}
}
```

Output:



TASK-3

a. Analyze how an effective Alumni Directory and Career Opportunities section can strengthen the relationship between the university and its alumni.

Answer:

An effective Alumni Directory and Career Opportunities section can strengthen the university-alumni relationship in several ways:

1. **Professional Networking:** Alumni can connect with professionals from different industries.
2. **Career Support:** Alumni can share job and internship opportunities.
3. **Mentorship:** Experienced alumni can guide younger graduates.
4. **University Engagement:** Alumni can remain connected with the university after graduation.

5. **Community Development:** Alumni can collaborate on different projects and events.
6. **Knowledge Sharing:** Experienced professionals can share their industry knowledge.

Therefore, the Alumni Directory and Career Opportunities section can create a strong professional community and maintain a long-term relationship between alumni and the university.

b. Compare the advantages and disadvantages of using CSS Flexbox and CSS Grid for designing the Alumni Directory section. Which one would you recommend and why?

Answer:

Feature	Flexbox	CSS Grid
Layout	One-dimensional	Two-dimensional
Structure	Works mainly with rows or columns	Works with rows and columns
Best for	Navigation bars and small components	Alumni Directory and card layouts
Main advantage	Simple and flexible	Better control over complex layouts
Disadvantage	Not ideal for complex 2D layouts	Slightly more difficult for beginners

So, I would recommend **CSS Grid** for the Alumni Directory because an Alumni Directory contains multiple profile cards arranged in **rows and columns**. Grid provides better control over the size, spacing, and position of these cards. However, Flexbox is useful for smaller components such as navigation bars and buttons. **Therefore, CSS Grid is more suitable for the Alumni Directory.**

End
Thank You